1) VALID URL

```
#include <iostream>
using namespace std;
int main(){
        string url;
        getline(cin,url);
        int start;
        bool valid = true;
        if(url.find("http://")){
                start = 7;
        }else if(url.find("https://")){
                start = 8;
        }else{
                valid = false;
        }
        if(valid&&url.length()<=start){
                valid = false;
        }if(valid&&url.find(' ')!= string::npos){
                valid = false;
        }
        int dot=url.find('.',start);
        if(dot == string::npos || dot == url.length()-1){
                valid = false;
        }
        if(valid){
                printf("Valid");
        }else{
                printf("Invalid");
        }
}
```

2) DATA MASKING (@ , LAST 4 NOS.)

```
#include<iostream>
using namespace std;
int main(){
        string s;
        cin >> s;
        int at = s.find('@');
        int n = s.length();
        string res="";
        if(at != string::npos){
                res+=s[0];
```

```
            for(int i=1;i<at;i++){
                res+='*';
            }res+=s.substr(at);
        }else{
            for(int i=0;i<n;i++){
                if(i>= n-4){
                    res+=s[i];
                }else if(s[i]=='-'){
                    res+=s[i];
                }else{
                    res+='*';
                }
            }
        }cout << res;
    }
```

3) ALPHA && DIGIT - PASS

```
#include<iostream>
using namespace std;
int main(){
    string s;
    cin >> s;
    bool valid=true;
    bool a = false;
    bool d = false;
    for(int i=0;i<s.length();i++){
        if(('A'<= s[i] && s[i] <= 'Z')||
            ('a'<= s[i] && s[i] <= 'z')){
                a=true;
            }
            if(('0'<= s[i]&& s[i] <= '9')){
            d=true;
        }
    }
    if(a&&d){
            cout << "True";
        }else{
            cout << "False";
        }
}
```

4) ANAGRAM

```
#include<iostream>
#include<algorithm>
using namespace std;
int main(){
        string s1,s2;
        cin >> s1 >> s2;
        if(s1.length() != s2.length()){
                cout << "False";
                return 0;
        }
        sort(s1.begin() , s1.end());
        sort(s2.begin() , s2.end());
        if(s1==s2) cout << "True";
        else cout << "False";
}
```

5) STRING EXPANSION

```
#include<iostream>
#include<cctype>
using namespace std;
int main(){
        string s;
        cin >> s;
        string res;
        for(int i=0;i<s.length();i++){
                if(i+1 < s.length() && isdigit(s[i+1])){
                        int count = s[i+1] - '0';
                        for(int j=0;j<count;j++){
                                res+=s[i];
                        } i++;
                }else{
                        res+=s[i];
                }
        } cout << res;
}
```

6) STRING COMPRESSION

```
#include<iostream>
using namespace std;
int main(){
```

```
        string s;
        cin >> s;
        string text ="";
        text += s[0];
        for(int i=1;i<s.length();i++){
                if(s[i]!=s[i-1])text+=s[i];
        }cout << text;
}
```

7) FIRST NON REPEATING CHARACTER

```
#include<iostream>
using namespace std;
int main(){
        string s;
        cin >> s;
        int count=0;
        for(int i=0;i<s.length();i++){
                count =0;
                for(int j=i;j<s.length();j++){
                        if(s[i]==s[j]){
                                count++;
                        }
                } if(count ==1 ){
                        cout << s[i];
                        return 0;
                }
        }
}
```

8) MIRROR WORD

```
#include<iostream>
#include<algorithm>
using namespace std;
int main(){
        string s1,s2;
        cin >> s1 >> s2;
        if(s1.length()!=s2.length()) cout << "false";
        reverse(s1.begin(),s1.end());
        if(s1 == s2) cout << "true";
        else cout << "false";
}
```

(OR)

```
for(int i=0;i<s1.length();i++){
        if(s1[i]!=s2[s1.length()-i-1]){
            cout << "false";
            return 0;
        }
```

9) PALINDROME

```
#include<iostream>
#include<algorithm>
using namespace std;
int main(){
    string s1,s2;
    cin >> s1;
    s2 = s1;
    reverse(s2.begin(),s2.end());
    if(s1==s2 )cout << "Valid";
}
```

10) LONGEST SUBSTRING PALINDROME

```
#include<iostream>
using namespace std;
int main(){
    string s;
    cin >> s;
    string longest = "";
    for(int i=0;i<s.length();i++){
        for(int j=i;j<s.length();j++){
            bool valid = true;
            int left = i , right = j;
            while(left<right){
                if(s[left]!=s[right]){
                    valid = false;
                    break;
                }
                left++;
                right--;
            }
            if(valid && j-i+1 > longest.length()){
                longest = s.substr(i,j-i+1);
            }
```

```
        }
    }
    cout << longest;
}
```

11) CASE CONVERSION

```
#include<iostream>
using namespace std;
int main(){
    string s;
    cin >> s;
    for(int i=0;i<s.length();i++){
        if('a'<= s[i] && s[i]<= 'z'){
            s[i]=s[i]-32;
        }else if ('A'<=s[i]&&s[i]<='Z'){
            s[i]=s[i]+32;
        }
    }cout << s;
}
```

12) VALID IPV4 IPV6

```
#include <iostream>
#include <string>
#include <sstream>
using namespace std;

int main(){
    string ip;
    cin >> ip;

    if(ip.find('.') != string::npos){

        int count = 0;
        string part = "";
        bool valid = true;

        for(int i=0; i<=ip.length(); i++){

            if(i==ip.length() || ip[i]=='.'){

                if(part.empty()) valid = false;
```

```
            // check digits
            for(int j=0; j<part.length(); j++){
                if(!isdigit(part[j])) valid = false;
            }

            // convert string to int
            if(!part.empty()){
                int num;
                stringstream ss(part);
                ss >> num;

                if(num < 0 || num > 255) valid = false;
            }

            count++;
            part="";
        }
        else{
            part += ip[i];
        }
    }

    if(valid && count==4)
        cout << "Valid IPv4";
    else
        cout << "Invalid";
}

return 0;
}
```

13) PASSWORD VALIDATOR

```
#include<iostream>
#include<cctype>
using namespace std;
int main(){
        string s;
        cin >> s;
        bool dig,low,up,spc = false;
        for(int i=0;i<s.length();i++){
                if(isdigit(s[i])) dig = true;
                else if (islower(s[i])) low = true;
```

```
            else if (isupper(s[i])) up = true;
            else spc = true;
        }
        if( dig && low && up && spc) cout << "Valid";
        else cout << "Invalid";
}
```

14) LONGEST COMMON PREFIX

```
#include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        string s[n];
        for(int i=0;i<n;i++){
            cin >> s[i];
        } string prefix;
        for(int i=0;i<s[0].length();i++){
            char ch = s[0][i];
            bool valid = true;
            for(int j=0;j<n;j++){
            if(i >= s[j].length() || s[j][i]!= ch){
                valid = false;
                break;
            }
            }
            if(valid) prefix += ch;
            else break;
        }
        cout << prefix;
}
```

15) PANGRAM

```
#include<iostream>
#include<cctype>
using namespace std;
int main(){
        string s;
        getline(cin,s);
        bool seen[26] = {false};
        for(int i=0;i<s.length();i++){
            char c = tolower(s[i]);
```

```
            if('a'<=c&&c<='z'){
                seen[c-'a']=true;
            }
        }
        for(int i=0;i<26;i++){
            if(!seen[i]){
                cout << "False";
                return 0;
            }
        }
        cout << "True";
}
```

16) SUBSEQUENCE OF ANOTHER STRING?

```
#include<iostream>
using namespace std;
int main(){
        string s1 , s2;
        cin >> s1 >> s2;
        int i=0,j=0;
        while(i<s1.length()&&j<s2.length()){
            if(s1[i]==s2[j]){
                i++;
            }j++;
        }if(i==s1.length()){
            cout << "true";
        }else{
            cout << "False";
        }
}
```

17) CAMEL CASE

```
#include<iostream>
#include<cctype>
using namespace std;
int main(){
        string s;
        getline(cin , s);
        string res = "";
        bool cap = false ;
        for(int i=0;i<s.length();i++){
```

```
            if(s[i]==' '){
                cap = true;
            }else{
                if(cap){
                    res+= toupper(s[i]);
                    cap = false;
                }else{
                    res+=tolower(s[i]);
                }
            }
        }
        cout << res;
}
```

18) COUNT OF LETTERS IN BETWEEN

```
#include<iostream>
#include<sstream>
using namespace std;
int main(){
        string s;
        getline(cin , s);
        stringstream ss(s);
        string word;
        while(ss >> word){
            if(word.length()<=2){
                cout << word << " ";
            }else{
                cout << word[0] << word.length()-2 << word[word.length()-1]<< " ";
            }
        }
}
```

ARRAYS

1) ABSOLUTE DIFFERENCE

```
#include<iostream>
#include<cstdlib>
using namespace std;
int main(){
        string s;
        cin >> s;
```

```
        int count =0;
        for(int i=0;i<s.length()-1;i++){
                int d1 = s[i]-'0';
                int d2 = s[i+1]-'0';
                if(abs(d1-d2) >= 2) count++;
        }cout << count;
}
```

2) PIVOT INDEX

```
#include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        int total;
        int nums[n];
        for(int i=0;i<n;i++){
                cin >> nums[i];
                total+=nums[i];
        }int left = 0;
        for(int i=0;i<n;i++){
                int right = total - left - nums[i];
                if(left==right){
                        cout << i;
                        return 0;
                }
                left+=nums[i];
        }cout << -1;
}
```

3) MAXIMUM SUM W/O ADJ. ELEMENTS

```
 #include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        int num[n];
        for(int i=0;i<n;i++){
                cin >> num[i];
        }
        if(n==1){
                cout << num[0];
                return 0;
```

```
        }
        int prev2=num[0];
        int prev1=max(num[0],num[1]);
        for(int i=2;i<n;i++){
                int curr = max(prev1, prev2+num[i]);
                prev2=prev1;
                prev1=curr;
        }
        cout << prev1;
}
```

4) SUB ARITHMETIC ARRAY - MAX LENGTH

```
#include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        int num[n];
        for(int i=0;i<n;i++){
                cin >> num[i];
        }int curr = 2 , max =2;
        int diff = num[1]-num[0];

        for(int i=2;i<n;i++){
                if(num[i-1]-num[i] == diff){
                        curr++;
                }else{
                        curr=2;
                        diff = num[i-1]-num[i];
                }
                if( curr > max ) max = curr;
        }cout << max;
}
```

5) MAJORITY ELEMENT

```
class Solution {

public:

int majorityElement(vector<int>& nums) {

        int count=0,candidate;
```

```
        for(int i=0;i<nums.size();i++){

            if(count==0) candidate = nums[i];

            if(nums[i]==candidate) count++;

            else count--;

        } return candidate;

}

};
```

6) MOVE ZEROES

```
class Solution {
public:
    void moveZeroes(vector<int>& nums) {
        int j=0;
        for(int i=0;i<nums.size();i++){
            if(nums[i]!=0){
                swap(nums[i],nums[j]);
                j++;
            }
        }

    }
};
```

7) PRIME COUNT

```
#include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        int arr[n];
        for(int i=0;i<n;i++){
                cin >> arr[i];
        }int count = 0;
        for(int i=0;i<n;i++){
                int num = arr[i];
                bool prime = true;
```

```
                if(num <=1) prime=false;
                else{

                for(int j=2;j*j<=num;j++){
                        if(num%j==0){
                                prime=false;
                                break;
                        }
                }
        }if(prime){
                count++;
        }
        }cout << count;
}
```

8) FROM TARGET , REVERSE

```
#include<iostream>
using namespace std;
int main(){
        int n;
        cin >> n;
        int arr[n];
        for(int i=0;i<n;i++){
                cin >> arr[i];
        }
        int target;
        cin >> target;
        int start=target-1;
        for(int i=0;i<n;i++){
                if(i<start){
                        cout<<arr[i];
                }else{

                        for(int j=n-1;j>=start;j--){
                                cout<<arr[j] << " ";
                        }return 0;

        }
}
}
```

9) DIST. BTWN. 1’s

```
#include<iostream>
using namespace std;
int main(){
        string s;
        cin >> s;
        int count =0,max=0;
        for(int i=0;i<s.length();i++){
                if(s[i]=='1'){
                        for(int j=i+1;j<s.length();j++){
                                if(s[j]=='1'){
                                        count = j-i-1;
                                        if(count > max){
                                                max = count;
                                        }break;
                                }
                        }
                }
        } cout << max;
}
```